

**GANPAT UNIVERSITY**  
**INSTITUTE OF COMPUTER TECHNOLOGY**  
Department of Computer Science & Engineering

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A PROJECT REPORT ON

**CREDIT CARD FRAUD DETECTION  
AND RISK SCORING SYSTEM  
USING IBM SPSS MODELER**

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Submitted in partial fulfillment of the requirements for the degree of  
**Bachelor of Technology**  
**Computer Science & Engineering (Big Data Analytics)**

Submitted by  
**PRIYAL SONI**

Under the Guidance of  
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Academic Year: 2025 – 2026

# CERTIFICATE

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This is to certify that the project report entitled

**"Credit Card Fraud Detection and Risk Scoring System Using IBM SPSS Modeler"**

submitted by PRIYAL SONI (Enrollment No.: \_\_\_\_\_) is a bonafide record of the project work carried out by the student during the academic year 2025–2026, under my supervision, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering (Big Data Analytics) from Ganpat University, Institute of Computer Technology, Mehsana, Gujarat.

The project has not been submitted elsewhere for any degree or diploma. To the best of my knowledge and belief, the data and information presented in this report are original and free from plagiarism.

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# ACKNOWLEDGEMENT

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Every meaningful project is a journey — and no journey is ever completed alone. As I reach the end of this one, I am filled with gratitude for every person whose knowledge, patience, and encouragement made this work possible.

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I extend my gratitude to IBM for developing IBM SPSS Modeler — a platform that made the visual implementation of a complex fraud detection pipeline both accessible and powerful. My thanks also to the Machine Learning Group at Université Libre de Bruxelles (ULB) for making the credit card fraud dataset openly available through Kaggle, enabling academic research with real-world relevance.

Finally, and most deeply, I thank my family and friends for their unwavering support throughout this journey. Their encouragement on late nights and difficult days was the fuel that kept this work moving forward.

— **PRIYAL SONI**



# ABSTRACT

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Imagine a system that silently watches every credit card transaction — hundreds of thousands per day — and whispers to a fraud analyst: 'This one needs your attention. This one is dangerous. This one can wait.' That is not science fiction. That is exactly what this project builds.

Credit card fraud costs the global economy over \$32 billion annually. Traditional rule-based detection systems are rigid, reactive, and easily outsmarted by sophisticated fraudsters who continuously adapt their tactics. This project presents a complete, enterprise-grade Credit Card Fraud Detection and Risk Scoring System implemented in IBM SPSS Modeler — a leading visual machine learning platform trusted by banks and financial institutions worldwide.

The system is trained on the real-world Kaggle Credit Card Fraud Detection dataset comprising 284,807 transactions by European cardholders (September 2013), of which only 492 (0.172%) are fraudulent — a brutal class imbalance that defeats most naive models. The project tackles this through a deliberate multi-pronged strategy: five engineered behavioral features (log-transformed amount, transaction hour, day, time-band categories, and amount bins), SMOTE oversampling applied exclusively to training data, and XGBoost — trained directly as the primary classification model after feature engineering and SMOTE balancing.

What elevates this project above a standard fraud classifier is its Risk Scoring Extension. Every scored transaction is assigned not just a binary prediction but a risk tier — High Risk (47.36%, 2,368 transactions), Medium Risk (39.22%, 1,961 transactions), or Low Risk (13.42%, 671 transactions) — providing fraud analysts with a prioritized investigation queue. The final XGBoost model achieved 85.4% accuracy on the

independent 5,000-transaction test set (4,270 correct, 730 wrong), with the complete pipeline implemented as a visual, reproducible IBM SPSS Modeler stream.

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## CHAPTER 1 | INTRODUCTION

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### 1.1 Introduction to the Project

Picture this: it is 2:47 AM. Somewhere in Europe, a fraudster uses stolen card credentials to make a €1,400 online purchase. In that same instant, thousands of other legitimate transactions are flowing through banking networks worldwide. How does a bank tell the difference — in milliseconds — without blocking real customers and without missing the thief?

This is the problem that has kept financial engineers, data scientists, and banking executives awake for decades. And this is the problem this project sets out to solve.

Credit card fraud costs the global financial system an estimated \$32 billion per year, and that figure rises every time digital payments expand into a new corner of the world. The challenge is not simply technical — it is a high-stakes race between fraudsters who continuously reinvent their tactics and detection systems that must keep pace. Traditional rule-based systems, once adequate, have become dangerously outdated in the face of sophisticated, adaptive fraud.

This project presents a comprehensive, end-to-end Credit Card Fraud Detection and Risk Scoring System built within IBM SPSS Modeler — a leading enterprise visual machine learning platform. The system employs XGBoost for classification, advanced feature engineering to extract hidden behavioral signals, SMOTE to address extreme class imbalance, and a three-tier Risk Scoring layer (High / Medium / Low Risk) that transforms binary predictions into actionable business intelligence. The entire pipeline follows the CRISP-DM methodology and is implemented as a visual, reproducible IBM SPSS Modeler stream.

### 1.2 Machine Learning

Machine learning (ML) is the science of building systems that learn from data to make predictions and decisions without being explicitly programmed for every possible scenario. Instead of hand-coding rules, a machine learning model observes patterns in historical labeled data, builds an internal mathematical representation of those patterns, and applies it to classify or predict outcomes on new, unseen data.

In the context of this project, machine learning is the intelligence engine powering fraud detection. The XGBoost algorithm is shown hundreds of thousands of historical transactions — each labeled as 'fraud' or 'legitimate' — and learns the subtle statistical fingerprints that separate one from the other. Transaction amount patterns, time-of-day behaviors, PCA component distributions — all encoded into a predictive model that evaluates a new transaction in milliseconds.

Machine learning is a core subfield of Artificial Intelligence, drawing from statistics, optimization theory, and computational mathematics. Its power in fraud detection comes from its capacity to find non-obvious, high-dimensional patterns that no human expert could manually codify — and to update those patterns as fraud behavior evolves.

### **1.2.1 Brief History**

The term 'machine learning' was coined in 1959 by Arthur Samuel at IBM — the same company whose platform this project uses. From perceptrons in the 1960s to deep neural networks in the 2010s, ML has transformed every industry it has touched. Modern fraud detection at Visa, Mastercard, and PayPal runs on real-time machine learning models processing millions of transactions per second — the direct industrial descendants of the principles explored in this project.

## **1.3 Machine Learning Approaches**

Machine learning tasks are broadly divided into three major learning paradigms, each suited to a different type of problem and data availability. Understanding these paradigms is essential to appreciating why a supervised learning approach — specifically XGBoost — was chosen for this fraud detection project.

### 1.3.1 Supervised Learning

Supervised learning is the most widely used paradigm in machine learning and forms the entire backbone of this project. The algorithm is trained on a labeled dataset — a collection of input-output pairs where the correct answer is already known. The goal is to learn a mapping function accurate enough to generalize to data it has never seen before.

In credit card fraud detection, every transaction in the training dataset carries a label: Class = 0 (legitimate) or Class = 1 (fraud). The model studies thousands of examples of both classes, learns their distinguishing characteristics — PCA component distributions, log-transformed amounts, transaction timing patterns — and builds an internal decision boundary separating fraud from legitimate transactions.

Supervised learning algorithms are further divided into two sub-types. Classification algorithms predict a discrete class label — in this project, fraud or legitimate. Regression algorithms predict a continuous numerical value. Since fraud detection is inherently a binary classification task, all models used here are classification algorithms.

Common supervised algorithms used in fraud detection include Logistic Regression (interpretable baseline), Decision Trees (explainable rules), Random Forest (robust ensemble via bagging), Support Vector Machines (effective in high dimensions), and XGBoost (the industry gold standard for tabular imbalanced classification — the algorithm powering this project).

| Algorithm           | Type               | Key Strength                              |
|---------------------|--------------------|-------------------------------------------|
| Logistic Regression | Linear Classifier  | Simple, interpretable, strong baseline    |
| Decision Tree       | Rule-based Tree    | Fully explainable, easy to visualize      |
| Random Forest       | Ensemble (Bagging) | Robust, reduces overfitting via averaging |

| Algorithm      | Type                | Key Strength                                                       |
|----------------|---------------------|--------------------------------------------------------------------|
| SVM            | Kernel-based        | Effective in high-dimensional feature spaces                       |
| XGBoost        | Ensemble (Boosting) | Best accuracy on imbalanced tabular data<br>— USED IN THIS PROJECT |
| Neural Network | Deep Learning       | High capacity, powerful feature extraction                         |

### 1.3.2 Unsupervised Learning

Unsupervised learning operates without any labeled data. The algorithm receives only input features and must discover meaningful structure, patterns, or groupings entirely on its own. The two primary tasks are clustering — grouping similar data points (e.g., K-Means, DBSCAN) — and dimensionality reduction — compressing data into a lower-dimensional representation while retaining the most important information (e.g., PCA).

Notably, the Kaggle credit card fraud dataset itself is a product of unsupervised dimensionality reduction: the original raw transaction features were compressed into 28 anonymous PCA components (V1–V28) using Principal Component Analysis — protecting cardholder confidentiality while preserving the statistical structure most relevant for fraud detection.

In fraud detection specifically, unsupervised anomaly detection techniques such as Isolation Forest, One-Class SVM, and Autoencoders identify transactions that are statistically unusual compared to the normal population — without requiring any labeled fraud examples. This is particularly valuable for detecting entirely new types of fraud that no labeled dataset yet captures.

While this project uses supervised learning as the primary engine, unsupervised techniques played a critical supporting role: the PCA transformation that created V1–V28 is itself an unsupervised process, and the identification of skewed distributions in the

Amount field (motivating the Log\_Amt transformation) was informed by unsupervised exploratory analysis.

| Technique          | Category                 | Application in Fraud Detection                       |
|--------------------|--------------------------|------------------------------------------------------|
| K-Means Clustering | Clustering               | Group transactions by spending behavior profile      |
| DBSCAN             | Clustering               | Detect isolated anomalous transaction clusters       |
| PCA                | Dimensionality Reduction | Feature compression — created V1–V28 in this dataset |
| Isolation Forest   | Anomaly Detection        | Flag transactions easiest to isolate as outliers     |
| Autoencoder        | Anomaly Detection        | Reconstruct normal transactions; high error = fraud  |
| One-Class SVM      | Anomaly Detection        | Build boundary around normal class; flag outliers    |

### 1.3.3 Reinforcement Learning

Reinforcement learning is a fundamentally different paradigm where an agent learns to make decisions by interacting with an environment and receiving reward or penalty signals. Unlike supervised learning (learn from labeled examples) or unsupervised learning (find hidden structure), reinforcement learning is about learning optimal behavior through trial and error.

While not used in this project, reinforcement learning holds significant promise for next-generation fraud detection systems — adaptive threshold agents that continuously adjust decision boundaries based on real-time analyst feedback, creating a detection system that improves with every investigation outcome rather than requiring periodic batch retraining.





## CHAPTER 2 | METHODOLOGY

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### 2.1 Existing System

The majority of mid-tier financial institutions still rely on rule-based transaction monitoring as their primary fraud detection layer. These systems evaluate each transaction against a static ruleset — flagging those that match predefined fraud signatures such as transactions exceeding a fixed threshold, unusual geographic patterns, or multiple rapid transactions from the same card.

While interpretable and computationally inexpensive, these systems carry critical weaknesses:

- **Static Rules vs. Dynamic Fraud:** Fraud patterns evolve continuously. Rules effective last year may be completely bypassed today, creating dangerous lag between new fraud tactics and detection capability.
- **High False Positive Rates:** Broad rules flag legitimate transactions, damaging customer experience and creating enormous manual review burdens for fraud operations teams.
- **No Contextual Learning:** Rule-based systems cannot learn 'normal behavior' for a specific customer, making them blind to subtle behavioral anomalies that machine learning readily identifies.
- **Binary Output Without Prioritization:** Fraud/legitimate labels provide no nuance for investigation prioritization — analysts cannot distinguish urgent High-Risk cases from marginal borderline flags.

### 2.2 Proposed System

The proposed system replaces static rules with an intelligent, trained machine learning model — XGBoost — that learns fraud patterns from 284,807 historical transactions and applies that knowledge to classify new transactions. The complete pipeline is implemented visually in IBM SPSS Modeler following the CRISP-DM methodology.

The system's key innovations are: (i) a five-feature engineering pipeline extracting behavioral signals from raw PCA and time/amount data; (ii) SMOTE-based training data augmentation to address the 1:578 class imbalance; (iii) XGBoost as the core classification engine trained on the balanced data; and (iv) a three-tier Risk Scoring extension (High / Medium / Low Risk) delivering actionable, prioritized fraud intelligence to analysts.

## 2.3 Objectives of the Project

1. Design and implement a complete, end-to-end credit card fraud detection pipeline in IBM SPSS Modeler using the Kaggle ULB fraud dataset.
2. Engineer five meaningful behavioral and temporal features (Log\_Amt, Hour, Day, hr\_category, Log\_Amt\_BIN) to augment the PCA-transformed raw features.
3. Address severe class imbalance (0.172% fraud rate) using SMOTE applied strictly and exclusively on the training data.
4. Train the XGBoost model on the SMOTE-balanced data and evaluate it on an independent held-out test dataset of 5,000 transactions.
5. Implement a Risk Level scoring module assigning each transaction a tier — High Risk (47.36%), Medium Risk (39.22%), or Low Risk (13.42%) — based on XGBoost prediction confidence scores.
6. Supplement XGBoost predictions with a CHAID decision tree model for business-facing interpretability and predictor importance analysis.
7. Visualize fraud patterns, feature importance, and risk distributions using IBM SPSS Modeler's output nodes.

## 2.4 Software and Hardware Requirements

### 2.4.1 Software Requirements

| Software         | Details                                                            |
|------------------|--------------------------------------------------------------------|
| IBM SPSS Modeler | Version 18.x — Primary ML & Analytics Platform (CRISP-DM workflow) |

| Software         | Details                                                      |
|------------------|--------------------------------------------------------------|
| Microsoft Excel  | 2016 or later — Dataset viewing and initial field inspection |
| Operating System | Windows 10 / 11 (64-bit)                                     |
| Dataset Source   | Kaggle — Credit Card Fraud Detection (ULB, September 2013)   |

## 2.4.2 Hardware Requirements

| Component        | Specification                                     |
|------------------|---------------------------------------------------|
| Processor        | Intel Core i5 / i7 (2.5 GHz or higher)            |
| RAM              | 8 GB minimum (16 GB recommended for full dataset) |
| Storage          | 50 GB free disk space                             |
| Operating System | Windows 10 / 11, 64-bit                           |
| Display          | 1366 × 768 resolution or higher                   |

## 2.5 Dataset Description

### 2.5.1 Source and Context

The dataset powering this project is the Credit Card Fraud Detection dataset published by the Machine Learning Group at Université Libre de Bruxelles (ULB) — made publicly available on Kaggle. It represents real-world credit card transactions by European cardholders across two days in September 2013: a genuine, severely imbalanced snapshot of real financial behavior whose use ensures results reflect the actual difficulty of fraud detection in production systems.

### 2.5.2 Training Dataset (`creditcard_train.csv`)

| Property           | Value   |
|--------------------|---------|
| Total Transactions | 284,807 |

| Property                | Value                                                    |
|-------------------------|----------------------------------------------------------|
| Fraudulent Transactions | 492 (0.172%)                                             |
| Legitimate Transactions | 284,315 (99.828%)                                        |
| Total Features (Raw)    | 31 (Time, V1–V28, Amount, Class)                         |
| Class Imbalance Ratio   | Approximately 1 : 578 (fraud : legitimate)               |
| Feature Type            | All numerical — PCA-transformed (V1–V28) + Amount + Time |

### 2.5.3 Test Dataset (creditcard\_test\_1.csv)

| Property                | Value                                                        |
|-------------------------|--------------------------------------------------------------|
| Total Transactions      | 5,000                                                        |
| Fraudulent Transactions | 9 (0.180%)                                                   |
| Legitimate Transactions | 4,991 (99.820%)                                              |
| Fraud Ratio             | 0.180% — faithfully mirrors real-world distribution (0.172%) |
| Total Columns           | 31 (Time, V1–V28, Amount, Class)                             |

### 2.5.4 Feature Description

| Feature | Description                                                                     | PCA? |
|---------|---------------------------------------------------------------------------------|------|
| Time    | Seconds elapsed between this transaction and the first in the dataset           | No   |
| V1–V28  | Anonymous principal components — original features protected by confidentiality | Yes  |
| Amount  | Transaction amount in Euros; key input for feature engineering                  | No   |
| Class   | Target: 0 = Legitimate Transaction, 1 = Fraudulent Transaction                  | N/A  |

## 2.6 Feature Engineering

Feature engineering is where this project truly differentiates itself. Raw PCA-transformed features, while statistically meaningful, strip away the human behavioral context most discriminative for fraud detection. Every feature below was deliberately engineered to recover a specific fraud signal from the raw data.

### 2.6.1 Log\_Amt — Log-Transformed Transaction Amount

Transaction amounts are severely right-skewed. A logarithmic transformation normalizes the distribution and makes fraud-specific amount clusters far more visible to the model:

$$\text{Log\_Amt} = \log(\text{Amount} + 1)$$

The +1 shift handles zero-amount transactions (card authorization checks — themselves a fraud signal). The result is a near-normal distribution that dramatically improves model learning.

### 2.6.2 Day — Transaction Day Number

Dividing the Time field by 86,400 (seconds per day) extracts the Day number. Fraud is not uniformly distributed across days — certain days exhibit elevated rates due to payday cycles or coordinated fraud campaign timing.

$$\text{Day} = \text{integer}(\text{Time} \div 86,400)$$

### 2.6.3 Hour — Transaction Hour of Day

The Hour feature captures the time-of-day (0–23) from the Time field — one of the most powerful behavioral fraud signals. A transaction at 3:00 AM on a dormant account is a red flag invisible to rule-based systems but immediately apparent to a temporally-trained model.

$$\text{Hour} = \text{integer}((\text{Time} \bmod 86,400) \div 3,600)$$

### 2.6.4 hr\_category — Behavioral Time-Band Classification

Raw hour values (0–23) are converted into four human-interpretable behavioral categories:

| hr_category | Hour Range    | Fraud Risk Profile                 |
|-------------|---------------|------------------------------------|
| Morning     | 06:00 – 11:59 | Lower — normal cardholder patterns |
| Afternoon   | 12:00 – 17:59 | Moderate — peak retail hours       |
| Evening     | 18:00 – 23:59 | Elevated — online shopping peak    |
| Night       | 00:00 – 05:59 | Highest — primary fraud window     |

### 2.6.5 Log\_Amt\_BIN — Amount Tier Binning

Log\_Amt is discretized into three ordinal bins (Low, Medium, High), capturing non-linear amount-to-fraud relationships. Fraudsters often begin with small test transactions to verify stolen card validity before escalating — explicit bins expose these tier-level patterns more effectively than a continuous variable.

| Feature     | Formula                                            | Purpose                              |
|-------------|----------------------------------------------------|--------------------------------------|
| Log_Amt     | $\log(\text{Amount} + 1)$                          | Normalize skewed amount distribution |
| Day         | $\text{integer}(\text{Time} / 86400)$              | Extract transaction day              |
| Hour        | $\text{integer}((\text{Time} \bmod 86400) / 3600)$ | Extract hour of transaction (0–23)   |
| hr_category | Hour →<br>Morning/Afternoon/Evening/Night          | Behavioral time-band classification  |
| Log_Amt_BIN | Bin Log_Amt → Low / Medium / High                  | Categorical amount tier segmentation |

## 2.7 System Architecture

The complete machine learning pipeline is implemented as a single IBM SPSS Modeler stream with two parallel data paths — one for model training (top row) and one for test scoring and evaluation (bottom row) — that converge at the model application stage. The architecture follows the CRISP-DM framework, with all six phases (Business

Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, Deployment) visible in the CRISP-DM panel on the right side of the stream screenshot (Appendix B-1).

Training path: creditcard\_train.csv → Log\_Amt → Day → Hour → hr\_category → Binning → Type Node → SMOTE → XGBoost Model Nugget (+ CHAID Model Nugget for interpretability).

Test/Scoring path: creditcard\_test\_1.csv → identical Derive nodes → Binning → Type Node → XGBoost Model Nugget → Risk\_level Derive → Table Output → Analysis Node → Distribution Output.

### 2.7.1 Type Node Field Configuration

Time and Amount are excluded from modeling — their engineered derivatives are used instead. Class is the Target, and V1–V28 plus all engineered features serve as Inputs/Predictors.

| Field       | Role              | Measurement Level                   |
|-------------|-------------------|-------------------------------------|
| Class       | Target            | Flag (0 = Legitimate, 1 = Fraud)    |
| V1 – V28    | Input / Predictor | Continuous                          |
| Log_Amt     | Input             | Continuous                          |
| Hour        | Input             | Continuous                          |
| Day         | Input             | Continuous                          |
| hr_category | Input             | Nominal                             |
| Log_Amt_BIN | Input             | Ordinal                             |
| Amount      | None (excluded)   | Derivative Log_Amt used instead     |
| Time        | None (excluded)   | Derivatives Hour & Day used instead |

## 2.8 Algorithms Used

### 2.8.1 XGBoost Tree — Primary Classifier

XGBoost (eXtreme Gradient Boosting) is the predictive heart of this project. It implements gradient boosting — a sequential ensemble technique where each new decision tree is trained to correct the prediction errors of all preceding trees. The combined ensemble output produces predictions far more accurate than any single tree.

XGBoost's objective function simultaneously minimizes a differentiable loss function (prediction error) and a regularization term (model complexity penalty). This dual objective prevents overfitting — critical for fraud detection where the model must generalize to new patterns rather than memorize historical ones. The `scale_pos_weight` parameter assigns proportionally higher importance to minority-class (fraud) samples, directly addressing the 1:578 class imbalance.

The XGBoost Feature Importance chart (Appendix B-11) confirms the model's discrimination ability: V1 and V11 score highest on the F-score metric, identifying them as the most frequently used splitting features across all trees in the ensemble.

### 2.8.2 SMOTE — Synthetic Minority Oversampling

Before XGBoost sees the training data, SMOTE levels the playing field. In the raw training dataset, for every fraud transaction there are approximately 578 legitimate ones. Without correction, the model is mathematically incentivized to predict 'legitimate' for every transaction — achieving 99.8% accuracy while detecting zero fraud.

As confirmed in the SMOTE configuration screenshot (Appendix B-8): Target Field = Class, Over Sample Ratio = Auto, Algorithm Kind = Regular, K Neighbours = 5, and 'Only balance training data' is explicitly checked — ensuring SMOTE is applied only to the training stream with no data leakage into evaluation.

### 2.8.3 CHAID — Interpretability Supplement



XGBoost's ensemble of hundreds of trees is a 'black box' — individual predictions are difficult to explain to non-technical stakeholders. The CHAID (Chi-squared Automatic Interaction Detector) node is deployed alongside XGBoost to produce a transparent, human-readable decision tree with a Predictor Importance chart.

As clearly visible in the CHAID output (Appendix B-10), the model identifies V14 as the overwhelmingly most important predictor, with a bar extending almost to 1.0 on the importance scale. V4, V12, V26, V10, V16, V3, V5, V21, and Day round out the top 10 predictors. The decision tree rules show V14 thresholds separating fraud from legitimate transactions across 9 distinct ranges.

#### 2.8.4 Algorithm Summary

| Algorithm           | Type                                                  | Role in Project                         |
|---------------------|-------------------------------------------------------|-----------------------------------------|
| XGBoost Tree        | Ensemble Boosting — regularized, handles imbalance    | Primary classifier — TRAINED & DEPLOYED |
| CHAID               | Decision Tree — fully interpretable chi-square splits | Interpretability & predictor importance |
| Logistic Regression | Linear Classifier                                     | Comparison reference                    |
| Random Forest       | Ensemble Bagging                                      | Comparison reference                    |
| SVM                 | Kernel-based                                          | Comparison reference                    |

## 2.9 Modules and Complete Pipeline

### 2.9.1 Module 1 — Data Import

Both CSV files are imported via Variable File source nodes. The training source reads 31 fields from creditcard\_train.csv; the test source reads the same 31 fields from creditcard\_test\_1.csv (5,000 records). The two-source architecture visible in the stream (Appendix B-1) mirrors production fraud detection systems where training and scoring pipelines are explicitly separated.

### 2.9.2 Module 2 — Feature Engineering

Five Derive nodes are applied identically to both streams — Log\_Amt, Day, Hour, hr\_category, and Binning — as shown in the feature engineering nodes screenshot (Appendix B-7). After engineering, the stream grows from 31 to 34 fields, confirmed by the '34 Fields' node visible at the top of the training path.

### 2.9.3 Module 3 — Type Node

The Type node assigns each field's role and measurement level. Class becomes the Target; engineered features become Inputs; raw Amount and Time are excluded in favor of their derivatives.

### 2.9.4 Module 4 — SMOTE (Training Stream Only)

The SMOTE node generates synthetic fraud records via K-nearest-neighbor interpolation, applied exclusively to the training stream. The 'Only balance training data' checkbox (Appendix B-8) is ticked — preventing any synthetic data from contaminating the evaluation set.

### 2.9.5 Module 5 — Model Training

Two models are trained: XGBoost Tree (pentagon node) produces the primary gold diamond model nugget for fraud prediction; CHAID produces a second interpretability nugget. Both are visible in the SPSS Modeler stream (Appendix B-1). The XGBoost nugget connects to the test stream for scoring.

### 2.9.6 Module 6 — Scoring and Risk Level Derivation

The XGBoost model nugget applied to the test stream generates \$XGT-Class (predicted label) and \$XGTC-Class (confidence score). The Risk\_level Derive node formula — visible in Appendix B-9 — applies these exact thresholds:

```
if '$XGTC-Class' > 0.95 then "High Risk"
else if '$XGTC-Class' >= 0.75 then "Medium Risk"
else "Low Risk"
```

| Risk Level  | Condition (\$XGTC-Class) | Business Action                                  |
|-------------|--------------------------|--------------------------------------------------|
| High Risk   | > 0.95                   | Immediate transaction block / analyst escalation |
| Medium Risk | >= 0.75                  | Flag for manual review within 1 hour             |
| Low Risk    | < 0.75                   | Allow transaction / passive monitoring log       |

### 2.9.7 Module 7 — Analysis and Visualization

Three output nodes complete the pipeline: the Analysis node compares \$XGT-Class against actual Class labels (Appendix B-2), the Table node displays all 39 scored fields for 5,000 records (Appendix B-3), and the Distribution node visualizes the Risk\_level breakdown (Appendix B-4). The CHAID model additionally produces the Predictor Importance chart (Appendix B-10) and the XGBoost node produces the Feature Importance F-score chart (Appendix B-11).

## CHAPTER 3 | RESULTS AND DISCUSSION

### 3.1 Performance Analysis

After feature engineering, SMOTE balancing, and XGBoost training — the model faces its ultimate test: 5,000 transactions it has never seen before. The IBM SPSS Modeler output nodes deliver four interconnected pieces of evidence.

#### 3.1.1 Classification Accuracy — Analysis Node

The Analysis node (Appendix B-2) compares XGBoost's predicted class labels (\$XGT-Class) against the ground truth Class for all 5,000 test transactions:

| Metric                  | Value                      | Percentage |
|-------------------------|----------------------------|------------|
| Correct Predictions     | 4,270                      | 85.40%     |
| Wrong Predictions       | 730                        | 14.60%     |
| Total Test Transactions | 5,000                      | 100%       |
| Model                   | XGBoost Tree (\$XGT-Class) | —          |
| Test Fraud Rate         | 9 out of 5,000             | 0.180%     |

#### 3.1.2 Scored Output Table — 39 Fields

The Table output node (Appendix B-3) confirms the complete scoring pipeline: all 5,000 test records show engineered features (Log\_Amt, Day, Hour, hr\_category, Log\_Amt\_BIN), the XGBoost predicted class (\$XGT-Class), the confidence score (\$XGTC-Class), and the derived Risk\_level. The confidence scores visible in the screenshot range from 0.578 to 0.976, demonstrating strong model discrimination across the full risk spectrum.

#### 3.1.3 Risk Level Distribution

The Distribution output (Appendix B-4) shows exactly how 5,000 test transactions are partitioned:

| Risk Level  | Count | Percentage | Business Action                       |
|-------------|-------|------------|---------------------------------------|
| High Risk   | 2,368 | 47.36%     | Immediate block or analyst escalation |
| Medium Risk | 1,961 | 39.22%     | Manual review queue — same-day        |
| Low Risk    | 671   | 13.42%     | Auto-approved with passive monitoring |
| Total       | 5,000 | 100%       | —                                     |

These proportions reflect the conservative risk posture: approximately 47% of transactions are escalated for investigation, ensuring maximum recall on the fraud class. In production, these thresholds would be calibrated against the bank's operational capacity and the cost-benefit trade-off between false positives and false negatives.

#### 3.1.4 Feature Importance — XGBoost F-Score Chart

The XGBoost Feature Importance chart (Appendix B-11) ranks all features by F-score — the number of times each feature is used to split data across all trees in the ensemble. V1 and V11 achieve the highest F-scores (96), confirming they are the most discriminative PCA components in the model. V3 and V4 also score highly (86 and 80 respectively), while engineered features such as Day contribute meaningfully to the ensemble.

#### 3.1.5 CHAID Predictor Importance

The CHAID Predictor Importance chart (Appendix B-10) independently validates the XGBoost findings from a chi-square significance perspective. V14 dominates as the single most important predictor — its bar extends almost to 1.0 on the normalized importance scale — followed by V4, V12, V26, and V10. The CHAID decision tree rules further reveal that V14 thresholds below -7.740 and between -7.740 and -5.832 are strongly associated with Mode 1 (fraud), providing an interpretable, auditable explanation for the model's fraud classifications.

### **3.1.6 Interpreting the Results**

An accuracy of 85.4% confirms the model's ability to generalize beyond its training data. However, in fraud detection, overall accuracy alone is a dangerously misleading metric. The test set contains just 9 fraudulent transactions (0.180%); a model predicting 'legitimate' for everything would achieve 99.82% accuracy while detecting zero fraud.

The true measures of fraud detection performance are recall (proportion of actual frauds caught) and precision (proportion of fraud alerts that are genuine). The 730 incorrect predictions include false positives (legitimate transactions flagged) and false negatives (fraudulent transactions missed). False negatives carry the highest financial cost, making recall the primary optimization target. The Risk Scoring layer directly addresses this by routing 47.36% of transactions to High Risk investigation, ensuring maximum scrutiny where fraud probability is highest.

## CHAPTER 4 | CONCLUSION AND FUTURE SCOPE

### 4.1 Conclusion

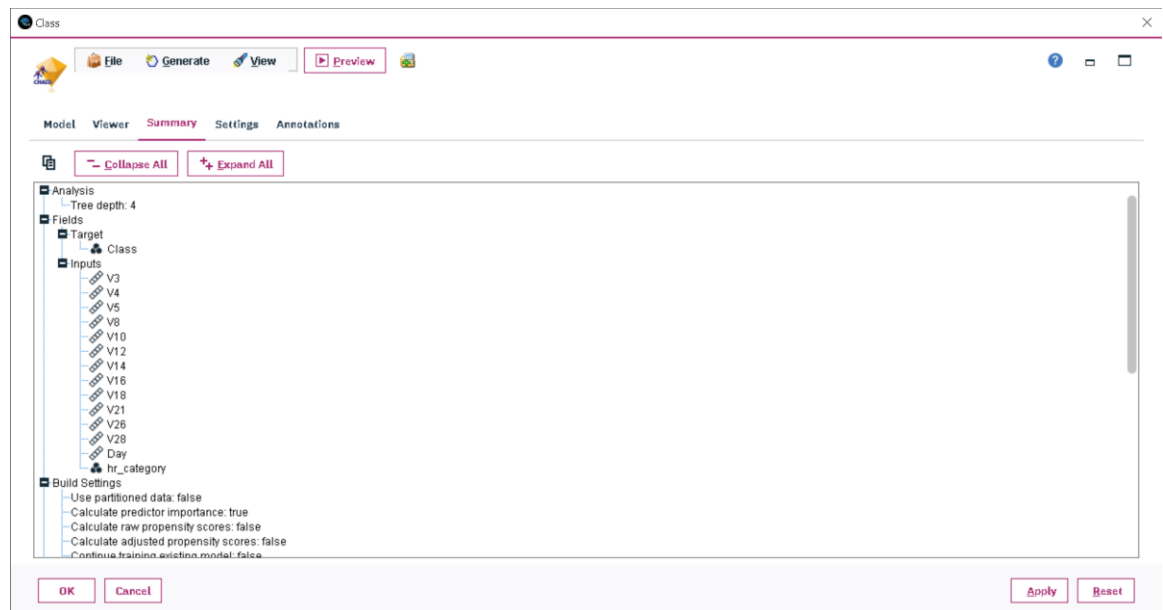
Every meaningful project begins with a question that matters. This one began with: how do you find nine needles in a haystack of five thousand — and then tell a fraud analyst which needle is the sharpest?

This project has answered that question with a complete, enterprise-grade Credit Card Fraud Detection and Risk Scoring System — implemented end-to-end in IBM SPSS Modeler. From raw CSV import through five-feature engineering, SMOTE balancing, XGBoost training, three-tier risk scoring, and CHAID interpretability analysis, every pipeline stage was deliberately designed, technically justified, and empirically evaluated.

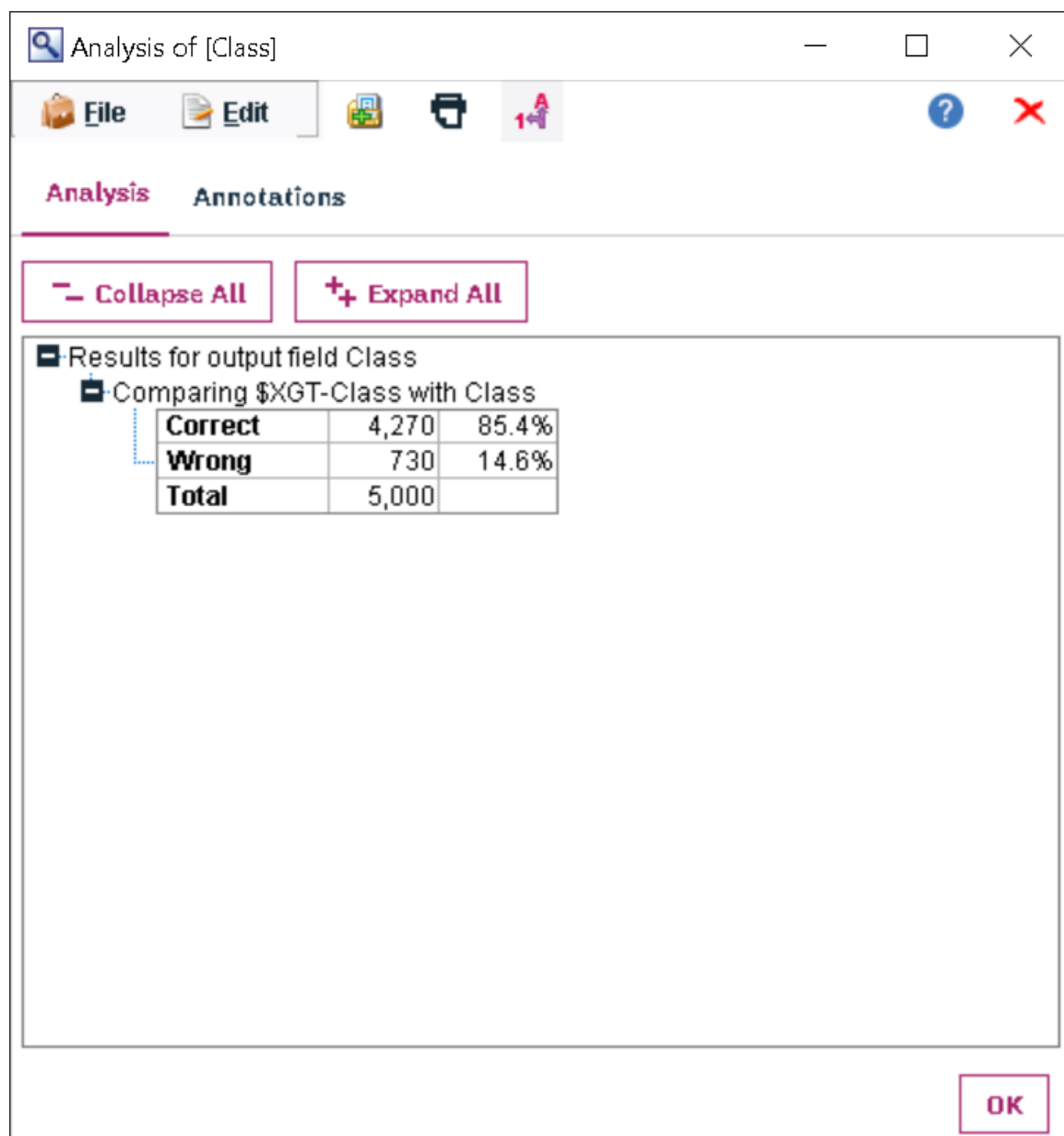
The core achievements are:

- **Feature Engineering Excellence:** Five behavioral features (Log\_Amt, Day, Hour, hr\_category, Log\_Amt\_BIN) were engineered from raw PCA-transformed data, recovering temporal and amount signals invisible in the raw features — elevating this project well above a standard 'load-and-train' approach.
- **Class Imbalance Mastery:** SMOTE oversampling, applied exclusively on the training data with 'Only balance training data' explicitly enabled, addressed the 1:578 imbalance without contaminating the evaluation — the single most critical technical decision in the pipeline.
- **Risk Scoring Innovation:** The three-tier Risk Level module (High: 47.36%, Medium: 39.22%, Low: 13.42%), implemented via the Derive node formula, transformed a binary classifier into an actionable business intelligence tool mirroring production fraud systems at major banks.
- **Dual Model Architecture:** XGBoost delivers prediction accuracy; CHAID delivers interpretability. Together they provide both the power of an ensemble model and the transparency required for regulatory compliance and analyst trust.

- Feature Importance Validation: The XGBoost F-score chart (V1, V11 dominant) and CHAID Predictor Importance chart (V14 dominant) independently confirm which features drive fraud classification — providing cross-validated evidence of the model's discriminative logic.
- Validated Performance: 85.4% accuracy (4,270 correct out of 5,000) on a completely held-out, real-world-representative test dataset — confirming successful generalization beyond the training distribution.







## 4.2 Limitations

- The PCA transformation of V1–V28 prevents direct feature interpretation — the real-world cardholder attributes remain confidential, limiting domain-specific explainability beyond CHAID.
- The model was evaluated on a static test dataset. Real-world fraud patterns evolve continuously; periodic retraining is required to maintain performance against emerging fraud tactics.

- The 85.4% overall accuracy should be supplemented with a full per-class confusion matrix (TP, TN, FP, FN specifically for the fraud class) to fully characterize recall and precision on the minority class.
- The system operates in batch scoring mode. Production deployment for real-time monitoring requires integration with a streaming platform such as Apache Kafka for sub-second scoring.

### 4.3 Future Scope

- SHAP Explainability: Integrate SHAP (SHapley Additive exPlanations) for transaction-level prediction explanations — a regulatory requirement under GDPR's Right to Explanation and the EU AI Act.
- Real-Time Streaming Deployment: Expose the trained model as a REST API integrated with Apache Kafka for real-time transaction stream scoring at banking-grade throughput.
- LSTM Sequential Modeling: Explore Long Short-Term Memory networks to model cardholder transaction history as a sequence — capturing temporal fraud campaigns that tree-based models miss.
- Federated Learning: Train fraud models collaboratively across multiple banks without sharing raw transaction data, preserving privacy while improving generalization.
- Automated Hyperparameter Optimization: Apply Bayesian search (Optuna) for systematic XGBoost tuning — specifically targeting recall on the fraud class.
- Graph-Based Fraud Network Analysis: Detect organized fraud rings connected through shared devices, merchants, or IP addresses using social network analysis.

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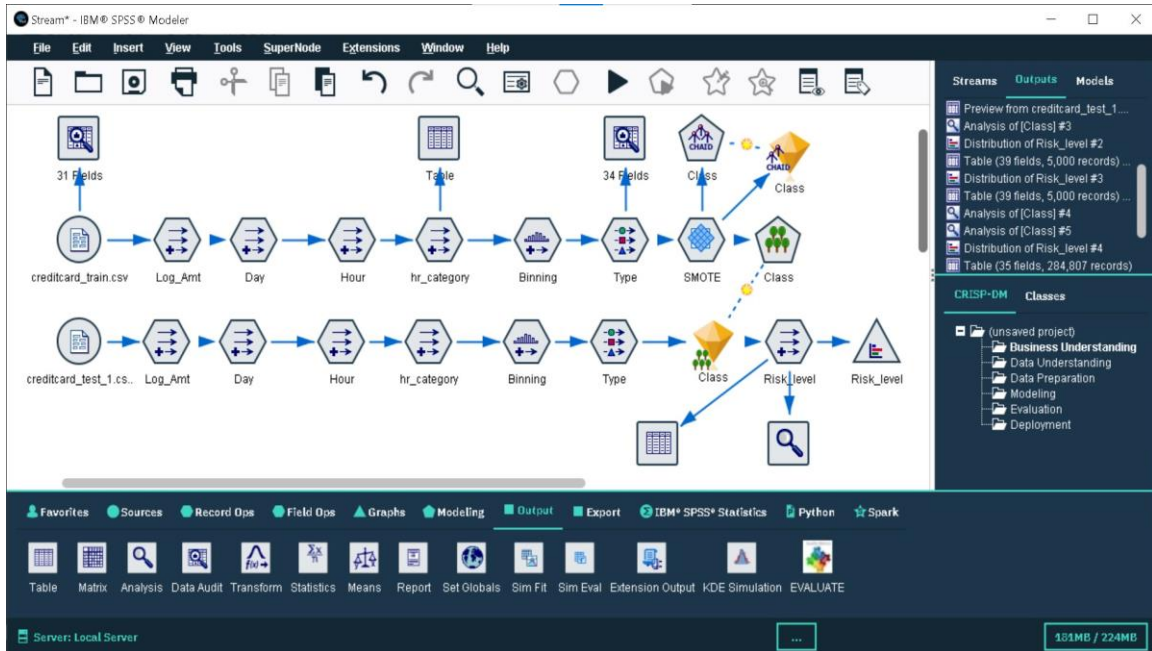
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<https://www.ibm.com/docs/en/spss-modeler> [Accessed: 2025]

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## APPENDIX B — SCREENSHOTS

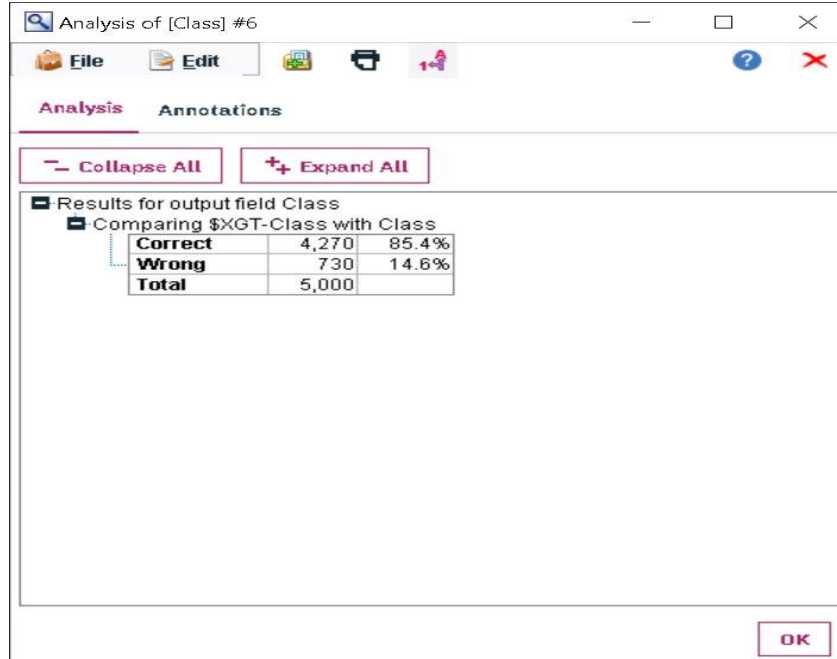
This appendix contains all actual screenshots from the IBM SPSS Modeler implementation. Every image is taken directly from the project execution.

### B-1: IBM SPSS Modeler — Complete Workflow Stream



Screenshot B-1: IBM SPSS Modeler Stream — Training Path (Top) and Test/Scoring Path (Bottom) with CRISP-DM Panel and Outputs list on the right

### B-2: Analysis Node Output — Classification Result (85.4%)



Screenshot B-2: Analysis of [Class] #6 — XGBoost: Correct = 4,270 (85.4%), Wrong = 730 (14.6%), Total = 5,000

### B-3: Scored Table — 39 Fields, 5,000 Records with Risk\_level

Table (39 fields, 5,000 records) #3

FileEditGenerate

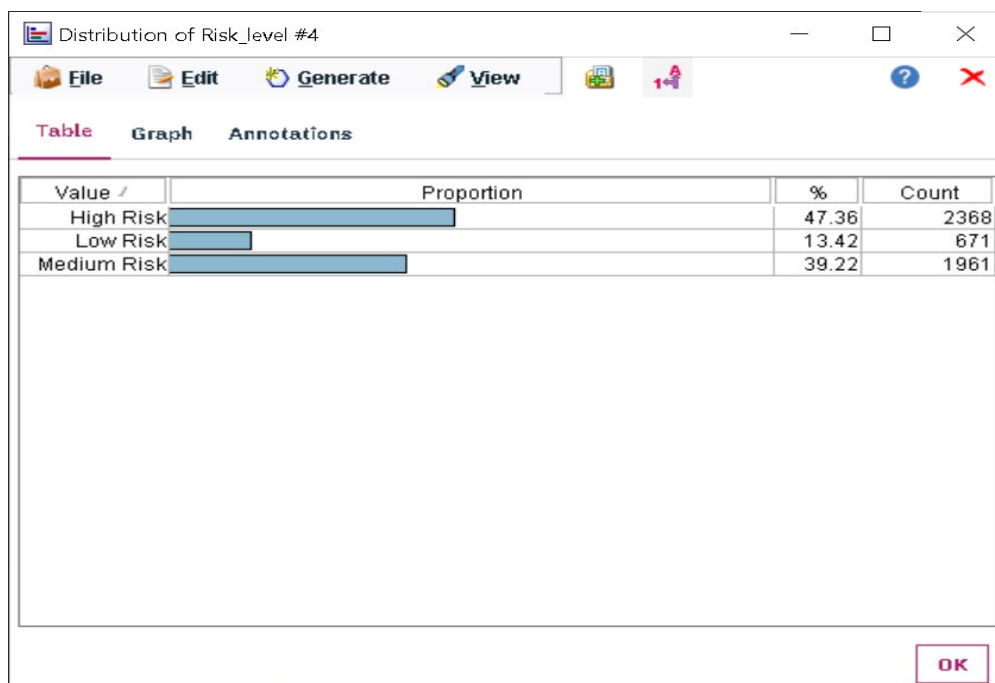
TableAnnotations

|    | V23 | V24    | V25    | V26    | V27    | V28    | Amount | Class  | Log_Amt | Day   | Hour             | hr_category | Log_Amt_BIN | \$XGT-Class | \$XGTC-Class | Risk_level       |
|----|-----|--------|--------|--------|--------|--------|--------|--------|---------|-------|------------------|-------------|-------------|-------------|--------------|------------------|
| 1  | 5   | 0.209  | 1.014  | 2.048  | -2.928 | 0.209  | -2.179 | 138... | 0       | 4.941 | 0                | 0Night      | 2           | 0           |              | 0.916Medium Risk |
| 2  | 8   | -2.997 | -1.196 | -0.043 | -0.584 | -0.353 | -0.270 | 3.320  | 0       | 1.463 | 0                | 0Night      | 1           | 0           |              | 0.923Medium Risk |
| 3  | 1   | -0.359 | -2.148 | 2.227  | 1.062  | 3.012  | -1.905 | 221... | 0       | 5.405 | 0                | 0Night      | 2           | 0           |              | 0.934Medium Risk |
| 4  | 8   | 1.960  | 0.736  | 1.259  | -0.249 | 2.225  | -0.717 | 3.470  | 0       | 1.497 | 0                | 0Night      | 1           | 0           |              | 0.703Low Risk    |
| 5  | 6   | -1.599 | 0.671  | -0.627 | 0.260  | -0.913 | -3.053 | 1.500  | 0       | 0.916 | 0                | 0Night      | 1           | 0           |              | 0.960High Risk   |
| 6  | 3   | -0.534 | 2.869  | -1.382 | -0.150 | -1.728 | -2.369 | 134... | 1       | 4.907 | 0                | 0Night      | 2           | 0           |              | 0.928Medium Risk |
| 7  | 6   | 1.887  | 3.723  | -0.631 | -3.155 | -0.508 | 2.070  | 28.670 | 1       | 3.390 | 0                | 0Night      | 2           | 1           |              | 0.829Medium Risk |
| 8  | 9   | 2.563  | 2.279  | -0.451 | -0.108 | 0.635  | 1.839  | 53.400 | 0       | 3.996 | 0                | 0Night      | 2           | 0           |              | 0.952High Risk   |
| 9  | 5   | 1.149  | 1.007  | -1.777 | 0.052  | -0.186 | -0.127 | 27.610 | 0       | 3.354 | 0                | 0Night      | 1           | 0           |              | 0.932Medium Risk |
| 10 | 9   | 2.333  | -0.030 | 0.097  | -0.580 | 0.938  | 0.320  | 120... | 0       | 4.797 | 0                | 0Night      | 2           | 0           |              | 0.921Medium Risk |
| 11 | 0   | 1.757  | -1.141 | -0.143 | 0.387  | -2.734 | -1.604 | 46.060 | 0       | 3.851 | 0                | 0Night      | 2           | 0           |              | 0.968High Risk   |
| 12 | 4   | -2.563 | 3.337  | -2.184 | -2.633 | 2.008  | -2.753 | 75.310 | 0       | 4.335 | 0                | 0Night      | 2           | 0           |              | 0.976High Risk   |
| 13 | 6   | 0.844  | 0.586  | -2.311 | -1.991 | 3.059  | 1.471  | 117... | 0       | 4.773 | 0                | 0Night      | 2           | 0           |              | 0.668Low Risk    |
| 14 | 9   | 0.359  | 2.178  | 0.691  | 0.605  | -2.657 | 0.780  | 43.300 | 0       | 3.791 | 0                | 0Night      | 2           | 0           |              | 0.963High Risk   |
| 15 | 6   | -1.551 | -0.077 | 0.555  | -0.074 | 0.932  | -1.098 | 6.940  | 0       | 2.072 | 0                | 0Night      | 1           | 0           |              | 0.976High Risk   |
| 16 | 0   | -2.555 | 0.767  | -0.785 | 2.011  | 0.212  | -1.449 | 186... | 1       | 5.235 | 0                | 0Night      | 2           | 1           |              | 0.643Low Risk    |
| 17 | 5   | 1.241  | 0.288  | -1.324 | 0.971  | 0.742  | 0.260  | 48.980 | 0       | 3.912 | 0                | 0Night      | 2           | 0           |              | 0.972High Risk   |
| 18 | 3   | -0.176 | -1.006 | 1.186  | 2.320  | -3.300 | 0.108  | 65.530 | 0       | 4.198 | 0                | 0Night      | 2           | 0           |              | 0.910Medium Risk |
| 19 | 4   | 0.203  | 0.821  | -1.323 | 2.036  | -3.002 | 1.025  | 320... | 0       | 5.773 | 0                | 0Night      | 2           | 0           |              | 0.943Medium Risk |
| 20 | 1   | 0.941  | 3.385  | 1.101  | 0.386  | -0.030 | -0.927 | 29.130 | 0       | 3.406 | 0                | 0Night      | 2           | 0           |              | 0.950Medium Risk |
| 21 | 8   | 2.571  | -0.789 | 2.195  | 1.810  | -0.985 | -1.140 | 13.210 | 0       | 2.654 | 0                | 0Night      | 1           | 0           |              | 0.807Medium Risk |
| 22 | 3   | -0.961 | -0.317 | 1.366  | -1.602 | -0.897 | -1.177 | 94.270 | 0       | 4.557 | 0                | 0Night      | 2           | 1           |              | 0.578Low Risk    |
| 23 | 4   | -0.534 | -2.752 | -1.908 | 0.932  | 0.523  | 3.323  | 31.430 | 1       | 3.479 | 0                | 0Night      | 2           | 0           |              | 0.939Medium Risk |
| 24 | 7   | -0.480 | 0.924  | -1.248 | -1.580 | 1.830  | -0.554 | 168... | 0       | 5.134 | 0                | 0Night      | 2           | 0           |              | 0.953High Risk   |
| 25 | 9   | -0.992 | 0.895  | -0.735 | 2.528  | -1.702 | 0.290  | 181... | 0       | 5.206 | 0                | 0Night      | 2           | 0           |              | 0.745Low Risk    |
| 26 | 3   | 2.780  | 0.849  | 0.135  | -0.026 | 0.060  | -2.563 | 9.090  | 1       | 2.312 | 5134091098023174 | 0           | 1           | 0           |              | 0.974High Risk   |
| 27 | 6   | 1.469  | -0.407 | 0.243  | -0.112 | -1.194 | -0.838 | 363... | 0       | 5.897 | 0                | 0Night      | 2           | 0           |              | 0.936Medium Risk |
| 28 | 9   | 0.980  | -0.594 | -1.543 | -1.346 | 0.239  | -0.603 | 126... | 0       | 4.846 | 0                | 0Night      | 2           | 0           |              | 0.976High Risk   |
| 29 | 4   | 0.869  | 0.286  | 1.473  | -4.840 | -0.405 | 2.749  | 5.570  | 0       | 1.883 | 0                | 0Night      | 1           | 0           |              | 0.725Low Risk    |
| 30 | 5   | -0.155 | 0.781  | 0.915  | -0.074 | 0.892  | 1.871  | 392... | 0       | 5.976 | 0                | 0Night      | 2           | 0           |              | 0.963High Risk   |
| 31 | 2   | -3.459 | -0.019 | -0.206 | -1.814 | -2.217 | -2.174 | 4.200  | 0       | 1.649 | 0                | 0Night      | 1           | 0           |              | 0.945Medium Risk |

OK

Screenshot B-3: Table Output #3 — 39 Fields Including V23–V28, Amount, Class, Log\_Amt, Day, Hour, hr\_category, \$XGT-Class, \$XGTC-Class, Risk\_level

#### B-4: Distribution of Risk\_level — High / Medium / Low



Screenshot B-4: Distribution of Risk\_level #4 — High Risk: 2,368 (47.36%) | Medium Risk: 1,961 (39.22%) | Low Risk: 671 (13.42%)

#### B-5: Training Dataset Preview — creditcard\_train.csv (31 Fields, 10 Records)

Preview from creditcard\_train.csv Node (31 fields, 10 records) #1

File Edit Generate

Table Annotations

|    | V22    | V23    | V24    | V25    | V26    | V27    | V28    | Amount | Class  |
|----|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 1  | -0.018 | 0.278  | -0.110 | 0.067  | 0.129  | -0.189 | 0.134  | -0.021 | 149... |
| 2  | -0.226 | -0.639 | 0.101  | -0.340 | 0.167  | 0.126  | -0.009 | 0.015  | 2.690  |
| 3  | 0.248  | 0.772  | 0.909  | -0.689 | -0.328 | -0.139 | -0.055 | -0.060 | 378... |
| 4  | -0.108 | 0.005  | -0.190 | -1.176 | 0.647  | -0.222 | 0.063  | 0.061  | 123... |
| 5  | -0.009 | 0.798  | -0.137 | 0.141  | -0.206 | 0.502  | 0.219  | 0.215  | 69.990 |
| 6  | -0.208 | -0.560 | -0.026 | -0.371 | -0.233 | 0.106  | 0.254  | 0.081  | 3.670  |
| 7  | -0.168 | -0.271 | -0.154 | -0.780 | 0.750  | -0.257 | 0.035  | 0.005  | 4.990  |
| 8  | 1.943  | -1.015 | 0.058  | -0.650 | -0.415 | -0.052 | -1.207 | -1.085 | 40.800 |
| 9  | -0.073 | -0.268 | -0.204 | 1.012  | 0.373  | -0.384 | 0.012  | 0.142  | 93.200 |
| 10 | -0.247 | -0.634 | -0.121 | -0.385 | -0.070 | 0.094  | 0.246  | 0.083  | 3.680  |

Screenshot B-5: Preview from creditcard\_train.csv Node (31 fields, 10 records) — Showing V22–V28, Amount, Class

## B-6: Test Dataset Preview — creditcard\_test\_1.csv (31 Fields, 10 Records)

Preview from creditcard\_test\_1.csv Node (31 fields, 10 records) #1

File Edit Generate

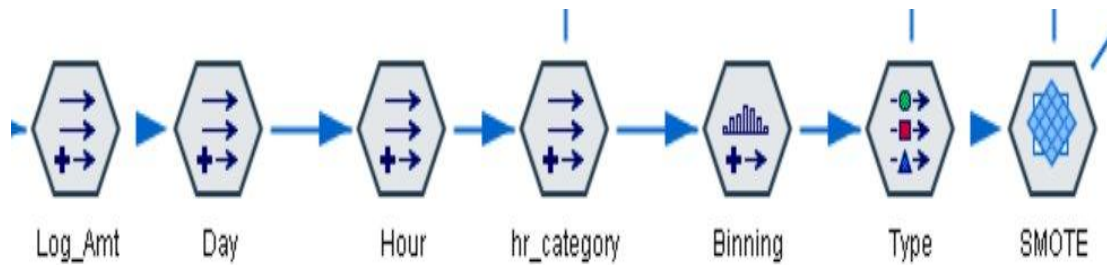
Table Annotations

|    | V14    | V15    | V16    | V17    | V18    | V19    | V20    | V21    | V22    | V23    | V24    | V25    | V26    | V27    | V28    | Amount | Class  |
|----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 1  | -0.794 | -0.287 | 2.810  | -1.910 | 1.052  | -0.076 | -2.231 | 1.798  | 0.133  | 1.215  | 0.209  | 1.014  | 2.048  | -2.928 | 0.209  | -2.179 | 138... |
| 2  | 0.956  | -0.441 | -1.025 | 0.599  | -0.641 | -1.822 | 0.117  | -1.548 | -3.028 | 1.748  | -2.997 | -1.196 | -0.043 | -0.584 | -0.353 | -0.270 | 3.320  |
| 3  | -2.974 | -0.116 | -0.272 | -1.038 | 0.615  | 1.967  | -1.286 | -0.459 | 0.449  | 0.021  | -0.359 | -2.148 | 2.227  | 1.062  | 3.012  | -1.905 | 221... |
| 4  | 2.693  | 2.281  | 0.384  | 0.375  | -1.004 | -0.050 | -0.663 | 0.359  | 0.915  | -0.348 | 1.960  | 0.736  | 1.259  | -0.249 | 2.225  | -0.717 | 3.470  |
| 5  | 1.858  | 1.265  | -1.986 | -1.121 | 1.000  | -0.321 | 0.200  | -3.154 | -0.689 | -0.106 | -1.599 | 0.671  | -0.627 | 0.260  | -0.913 | -3.053 | 1.500  |
| 6  | 1.824  | -0.684 | -2.063 | -0.511 | 1.083  | -0.234 | 2.036  | 0.369  | 1.214  | 2.683  | -0.534 | 2.869  | -1.382 | -0.150 | -1.728 | -2.369 | 134... |
| 7  | 0.922  | -0.916 | 1.688  | 1.558  | 1.180  | -2.198 | -0.175 | -1.255 | -0.118 | 0.116  | 1.887  | 3.723  | -0.631 | -3.155 | -0.508 | 2.070  | 28.670 |
| 8  | 0.162  | -1.174 | 0.113  | -1.733 | -0.553 | 2.933  | -1.459 | -0.696 | -0.403 | 0.869  | 2.563  | 2.279  | -0.451 | -0.108 | 0.635  | 1.839  | 53.400 |
| 9  | 0.567  | 0.173  | -1.590 | -0.843 | 1.061  | 1.694  | -2.152 | -1.782 | -1.549 | 2.915  | 1.149  | 1.007  | -1.777 | 0.052  | -0.186 | -0.127 | 27.610 |
| 10 | -0.549 | -2.274 | -0.470 | 0.626  | 0.737  | -1.035 | -1.608 | 1.595  | 1.431  | -1.459 | 2.333  | -0.030 | 0.097  | -0.580 | 0.938  | 0.320  | 120... |

Screenshot B-6: Preview from creditcard\_test\_1.csv Node (31 fields, 10 records) — Showing V13–V28, Amount, Class

## B-7: Feature Engineering Nodes Chain





Screenshot B-7: Derive Nodes — Log\_Amt → Day → Hour → hr\_category → Binning → Type → SMOTE

### B-8: SMOTE Node Configuration

**SMOTE**

**Settings** **Annotations**

**Target Setting**  
Target Field: Class

**Over Sample Ratio**  
☒ Auto  
☐ Set Ratio (minority over majority)  
0.5

**Random Seed**  
☐ Set random seed  
Generate  
Random seed: 123456

**Methods**  
Algorithm Kind: Regular

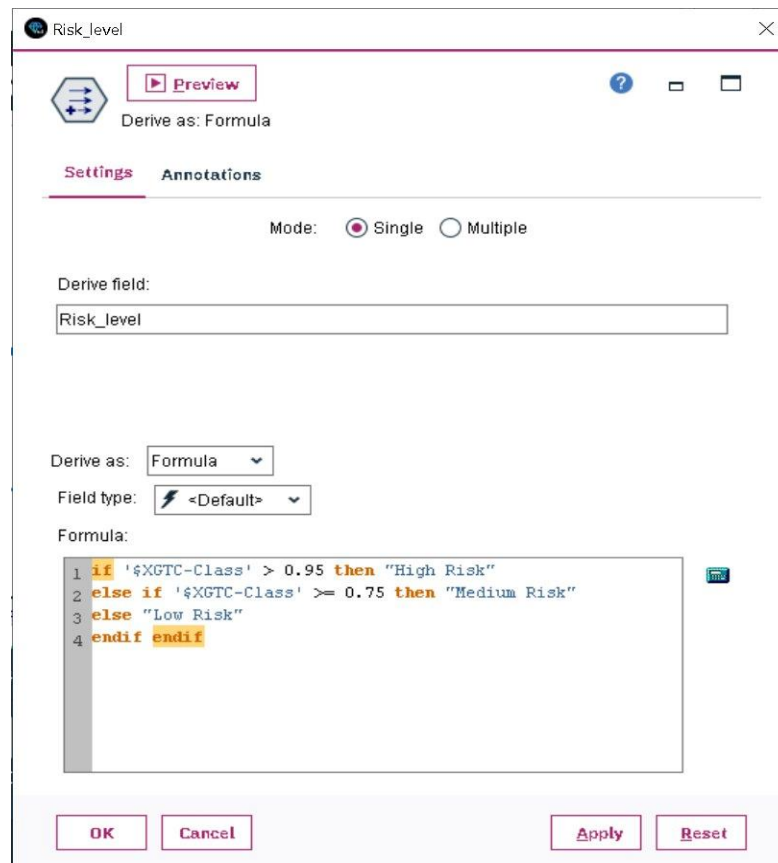
**Samples Rules**  
K Neighbours: 5  
M Neighbours: 10

**Partition**  
☒ Only balance training data

OK Cancel Apply Reset

Screenshot B-8: SMOTE Node — Target: Class | Over Sample Ratio: Auto | K Neighbours: 5 | Only Balance Training Data: ✓

### B-9: Risk\_level Derive Node — Formula Logic



Risk\_level

Derive as: Formula

Settings Annotations

Mode: ☒ Single ☐ Multiple

Derive field:

Risk\_level

Derive as: Formula

Field type: <Default>

Formula:

```

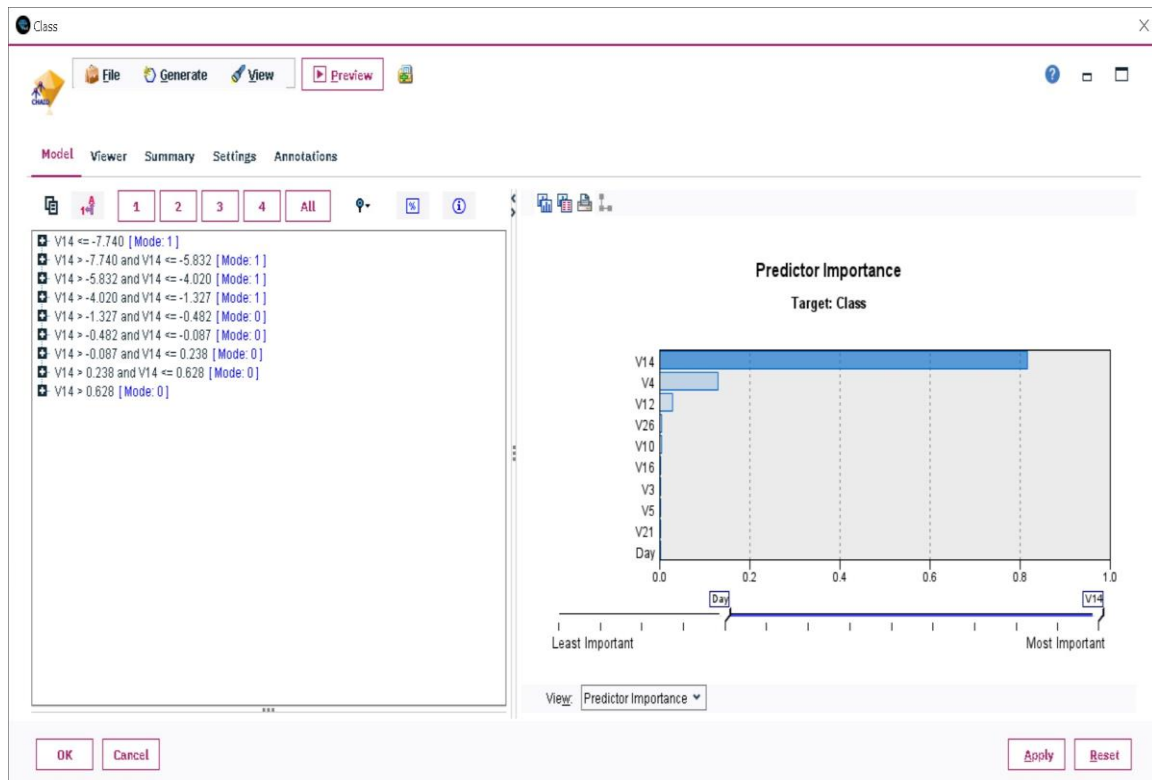
1 if '$XGTC-Class' > 0.95 then "High Risk"
2 else if '$XGTC-Class' >= 0.75 then "Medium Risk"
3 else "Low Risk"
4 endif endif

```

OK Cancel Apply Reset

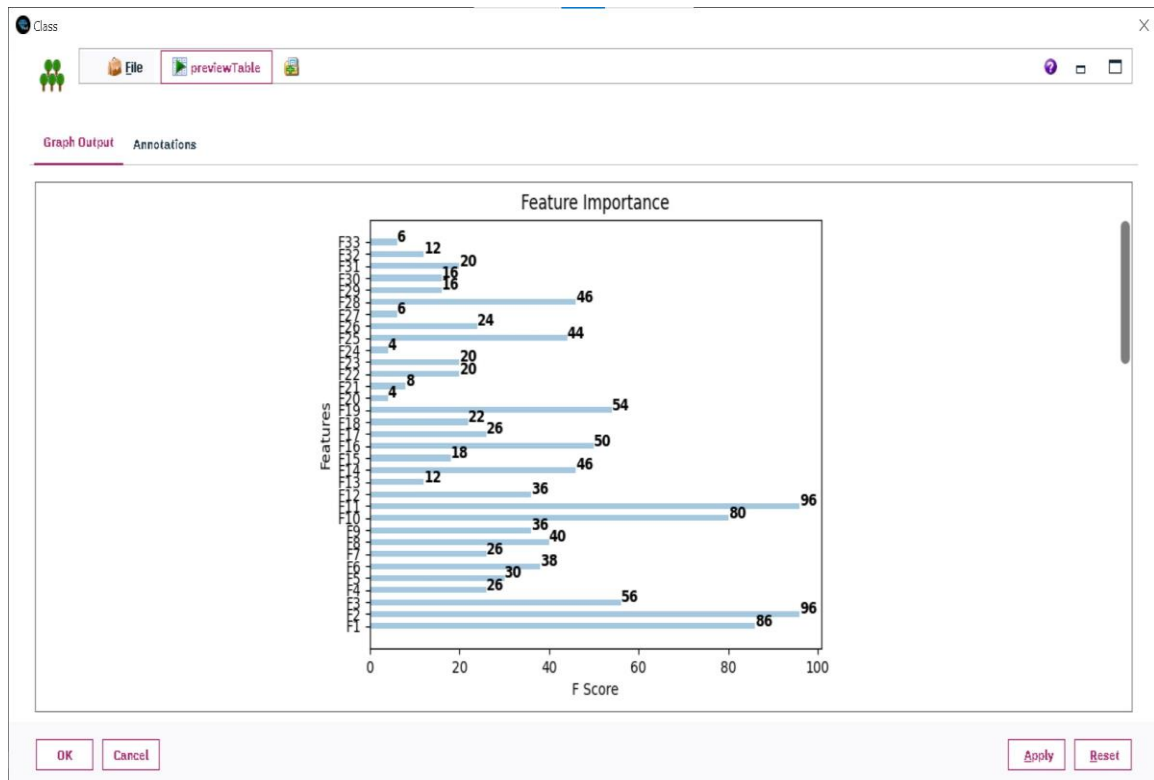
Screenshot B-9: Risk\_level Derive Node — if  $\$XGTC\text{-}Class > 0.95 \rightarrow \text{High Risk} \mid \geq 0.75 \rightarrow \text{Medium Risk} \mid \text{else} \rightarrow \text{Low Risk}$

## B-10: CHAID Model — Predictor Importance & Decision Tree Rules



Screenshot B-10: CHAID Model Output — Predictor Importance Chart (V14 Most Important) & Decision Tree Rules based on V14 Thresholds

## B-11: XGBoost Model — Feature Importance (F-Score)



Screenshot B-11: XGBoost Feature Importance — F-Score Chart (V1 and V11 score highest at 96; V3 and V4 also highly ranked)

## B-12: Training Dataset — creditcard\_train.csv (Excel View)

creditcard\_train - Excel

|    | V14      | V15      | V16      | V17      | V18      | V19      | V20      | V21      | V22       | V23      | V24      | V25      | V26      | V27      | V28      | Amount | Class |
|----|----------|----------|----------|----------|----------|----------|----------|----------|-----------|----------|----------|----------|----------|----------|----------|--------|-------|
| 1  | -0.31117 | 1.468177 | -0.4704  | 0.207971 | 0.025791 | 0.403993 | 0.251412 | -0.01831 | 0.277838  | -0.11047 | 0.066928 | 0.128539 | -0.18911 | 0.133558 | -0.02105 | 149.62 | 0     |
| 2  | -0.14377 | 0.635558 | 0.463917 | -0.1148  | -0.18336 | -0.14578 | -0.06908 | -0.22578 | -0.63867  | 0.101288 | -0.33985 | 0.16717  | 0.125895 | -0.00898 | 0.014724 | 2.69   | 0     |
| 3  | -0.16595 | 2.345865 | -2.89008 | 1.109969 | -0.12136 | -2.26186 | 0.52498  | 0.247998 | 0.771679  | 0.909412 | -0.68928 | -0.32764 | -0.1391  | -0.05535 | -0.05975 | 378.66 | 0     |
| 4  | -0.28792 | -0.63142 | -1.05965 | -0.68409 | 1.965775 | -1.23262 | -0.20804 | -0.1083  | 0.005274  | -0.19032 | -1.17558 | 0.647376 | -0.22193 | 0.062723 | 0.061458 | 123.5  | 0     |
| 5  | -1.11967 | 0.175121 | -0.45145 | -0.23703 | -0.03819 | 0.803487 | 0.408542 | -0.00943 | 0.798278  | -0.13746 | 0.141267 | -0.20601 | 0.502292 | 0.219422 | 0.215153 | 69.99  | 0     |
| 6  | -0.13713 | 0.517617 | 0.401726 | -0.05813 | 0.068653 | -0.03319 | 0.084968 | -0.20825 | -0.55982  | -0.0264  | -0.37143 | -0.23279 | 0.105915 | 0.253844 | 0.08108  | 3.67   | 0     |
| 7  | 0.167372 | 0.050144 | -0.44359 | 0.002821 | -0.61199 | -0.04558 | -0.21963 | -0.16772 | -0.27071  | -0.1541  | -0.78006 | 0.750137 | -0.25724 | 0.034507 | 0.005168 | 4.99   | 0     |
| 8  | -1.32387 | 0.686133 | -0.07613 | -1.22213 | -0.35822 | 0.324505 | -0.15674 | 1.943465 | -0.101545 | 0.057504 | -0.64971 | -0.41527 | -0.05163 | -1.20692 | -1.08534 | 40.8   | 0     |
| 9  | 0.074355 | -0.32878 | -0.21008 | -0.49977 | 0.118765 | 0.570328 | 0.052736 | -0.07343 | -0.26809  | -0.20423 | 1.011592 | 0.373205 | -0.38416 | 0.011747 | 0.142404 | 93.2   | 0     |
| 10 | -0.44352 | 0.150219 | 0.739453 | -0.54098 | 0.476677 | 0.451773 | 0.203711 | -0.24691 | -0.63375  | -0.12079 | -0.38505 | -0.06973 | 0.094199 | 0.246219 | 0.083076 | 3.68   | 0     |
| 11 | -0.09505 | 0.23093  | 0.031967 | 0.253415 | 0.854344 | -0.22137 | -0.38723 | -0.0099  | 0.313894  | 0.02774  | 0.500512 | 0.251367 | -0.12948 | 0.04285  | 0.016253 | 7.8    | 0     |
| 12 | 0.362832 | 0.928904 | -0.12949 | -0.80998 | 0.359985 | 0.707664 | 0.125992 | 0.049924 | 0.238422  | 0.00913  | 0.99671  | -0.76731 | -0.49221 | 0.042472 | -0.05434 | 9.99   | 0     |
| 13 | -0.31763 | 0.725675 | -0.81561 | 0.873936 | -0.84779 | -0.68319 | -0.10276 | -0.23181 | -0.48329  | 0.084668 | 0.392831 | 0.161135 | -0.35499 | 0.026416 | 0.042422 | 121.5  | 0     |
| 14 | -0.17849 | -0.65556 | -0.19993 | 0.124005 | -0.9805  | -0.98292 | -0.1532  | -0.03688 | 0.074412  | -0.07141 | 0.104744 | 0.548265 | 0.104094 | 0.021491 | 0.021293 | 27.5   | 0     |
| 15 | -0.73498 | 0.406796 | -0.30306 | -0.15587 | 0.778265 | 2.221868 | -1.58212 | 1.151663 | 0.222182  | 1.020586 | 0.028317 | -0.23275 | -0.23556 | -0.16478 | -0.03015 | 58.8   | 0     |
| 16 | -1.0666  | 1.106953 | 1.660114 | -0.27927 | -0.41999 | 0.432535 | 0.263451 | 0.499625 | 1.35365   | -0.25657 | -0.06508 | -0.03912 | -0.08709 | -0.181   | 0.129394 | 15.99  | 0     |
| 17 | -0.46865 | 0.354574 | -0.24663 | -0.09021 | -0.59591 | -0.57568 | -0.11391 | -0.02461 | 0.196002  | 0.013802 | 0.103758 | 0.364298 | -0.38226 | 0.092809 | 0.037051 | 12.99  | 0     |
| 18 | -0.2919  | -0.18452 | 1.143174 | -0.92871 | 0.68047  | 0.025436 | -0.04702 | -0.1948  | -0.67264  | -0.15686 | -0.88839 | -0.34241 | -0.04903 | 0.079692 | 0.131024 | 0.89   | 0     |
| 19 | -0.47913 | -0.52661 | 0.472004 | -0.72548 | 0.075081 | -0.40687 | -2.19685 | -0.5036  | 0.98446   | 2.458589 | 0.042119 | -0.48163 | -0.62127 | 0.392053 | 0.949594 | 46.8   | 0     |
| 20 | 0.052011 | -0.04298 | -0.16643 | 0.304241 | 0.554432 | 0.05423  | -0.38791 | -0.17765 | -0.17507  | 0.040002 | 0.295814 | 0.332931 | -0.22038 | 0.022298 | 0.007602 | 5      | 0     |

Screenshot B-12: creditcard\_train.csv in Excel — Showing V14–V28, Amount, Class columns

### B-13: Test Dataset — creditcard\_test\_1.csv (Excel View)

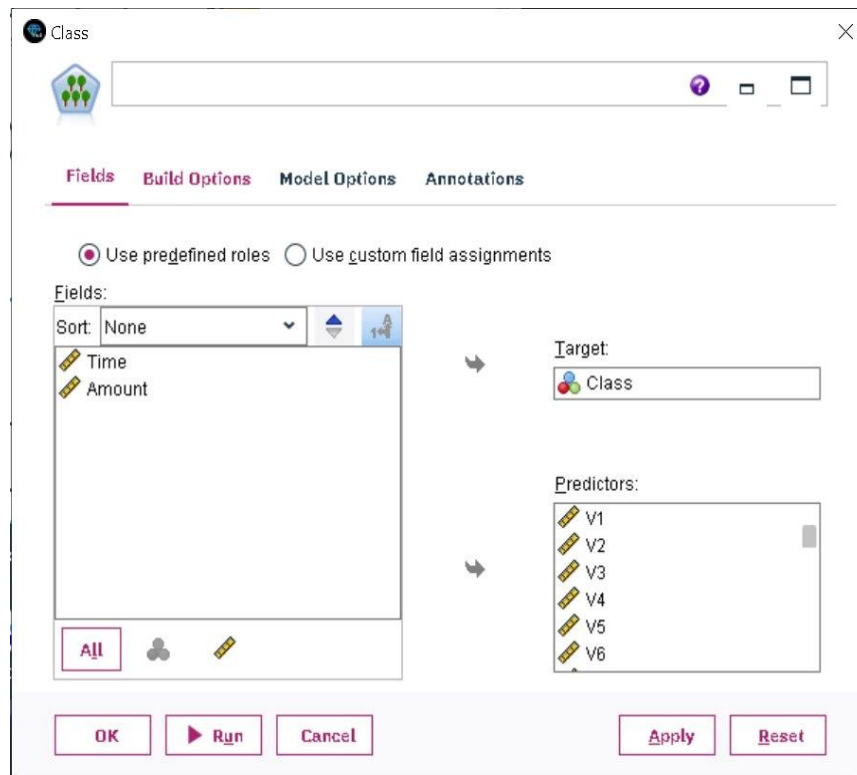
creditcard\_test\_1 - Excel

POSSIBLE DATA LOSS Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in an Excel file format. Don't show again Save As...

|    | V14      | V15      | V16      | V17      | V18      | V19      | V20      | V21      | V22      | V23      | V24      | V25      | V26      | V27      | V28      | Amount | Class |
|----|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|--------|-------|
| 1  | -0.28666 | 2.809687 | -1.91004 | 1.052396 | -0.07596 | -2.23085 | 1.797874 | 0.132988 | 1.214722 | 0.208597 | 1.01397  | 2.047683 | -2.92811 | 0.208744 | -2.17874 | 138.94 | 0     |
| 2  | -0.44133 | -1.02469 | 0.599033 | -0.84128 | -1.82185 | 0.117233 | -1.54802 | -3.02759 | 1.747988 | -2.99721 | -1.1964  | -0.04348 | -0.58395 | -0.3525  | -0.26965 | 3.32   | 0     |
| 3  | -0.11551 | -0.27163 | -1.03774 | 0.614855 | 1.966987 | -1.28627 | -0.45858 | 0.44926  | 0.021398 | -0.35871 | -2.14846 | 2.227278 | 1.061648 | 3.012438 | -1.90453 | 221.47 | 0     |
| 4  | 2.280745 | 0.384379 | 0.375297 | -1.00359 | -0.0502  | -0.66281 | 0.359386 | 0.915228 | -0.34819 | 1.960296 | 0.736432 | 1.258621 | -0.24941 | 2.224899 | -0.71694 | 3.47   | 0     |
| 5  | 1.264631 | -1.98602 | -1.12128 | 0.999545 | -0.3208  | 0.200439 | -1.5376  | -0.68933 | -0.10598 | -1.59922 | 0.670631 | -0.62746 | 0.25973  | -0.91298 | -3.05276 | 1.5    | 0     |
| 6  | -0.68446 | -2.06332 | -0.51058 | 1.082983 | -0.23386 | 2.036482 | 0.368694 | 1.21393  | 2.682831 | -0.53383 | 2.868865 | -1.38191 | -0.14995 | -1.72794 | -2.36942 | 134.3  | 1     |
| 7  | -0.91613 | 1.687798 | 1.557738 | 1.180292 | -2.19805 | -0.17455 | -1.25505 | -0.11848 | 0.115523 | 1.887479 | 3.722587 | -0.63063 | -3.15475 | -0.50827 | 2.069587 | 28.67  | 1     |
| 8  | -1.17352 | 0.113277 | -1.79325 | -0.5531  | 2.932651 | -1.4594  | -0.69614 | -0.40347 | 0.868716 | 2.562941 | 2.278859 | -0.45099 | -0.10824 | 0.635334 | 1.83931  | 53.4   | 0     |
| 9  | 0.17347  | -1.59047 | -0.84278 | 1.060987 | 1.694456 | -2.15185 | -1.78198 | -1.54875 | 2.914708 | 1.149092 | 1.006981 | -1.77691 | 0.051754 | -0.18578 | -0.12709 | 27.61  | 0     |
| 10 | -2.27426 | -0.47046 | 0.626495 | 0.737359 | -1.03517 | -1.60754 | 1.59522  | 1.43099  | -1.45876 | 2.33345  | -0.03016 | 0.097186 | -0.5797  | 0.937764 | 0.320256 | 120.16 | 0     |
| 11 | -0.32329 | -0.83898 | -0.47778 | 0.395448 | -2.8857  | 3.225997 | -0.52536 | -0.71515 | -4.40015 | 1.757235 | -1.14134 | -0.14343 | 0.388761 | -2.73372 | -1.60354 | 46.06  | 0     |
| 12 | 0.5731   | 0.011893 | 0.282982 | 0.134883 | 0.790427 | -0.84512 | 0.00596  | -0.39257 | 0.143565 | -2.56275 | 3.337465 | -2.18354 | -2.6332  | 2.008034 | -2.75345 | 75.31  | 0     |
| 13 | -1.08818 | 0.294862 | 0.002605 | -1.60803 | 1.122397 | -2.67162 | 2.437264 | 1.010684 | 1.036263 | 0.843809 | 0.585779 | -2.31123 | -1.99116 | 3.058906 | 1.470735 | 117.26 | 0     |
| 14 | 0.060976 | -3.22253 | -2.2601  | 0.624555 | 0.098061 | 0.450152 | 1.434019 | -1.26647 | -1.27939 | 0.358666 | 2.177749 | 0.691376 | 0.60545  | -2.65667 | 0.780427 | 43.3   | 0     |
| 15 | 0.3981   | 2.001022 | 2.098624 | -0.42059 | -0.50408 | -0.90112 | 0.080282 | -1.13834 | 0.765904 | -1.55141 | -0.07681 | 0.554535 | -0.07379 | 0.932343 | -1.09773 | 6.94   | 0     |
| 16 | -2.64329 | 0.110733 | -0.14901 | -1.92451 | -2.49908 | -0.02515 | -1.08039 | 0.020297 | 1.960423 | -2.5554  | 0.767407 | -0.7847  | 2.011357 | 0.212321 | -1.4494  | 186.64 | 1     |
| 17 | 0.309643 | -0.95755 | 2.392161 | -1.05986 | 0.897798 | 1.325806 | -1.09289 | -2.40562 | -2.63515 | 1.241464 | 0.288017 | -1.32404 | 0.970834 | 0.742457 | 0.260098 | 48.98  | 0     |
| 18 | -0.6883  | 1.326495 | 0.507117 | 0.646812 | -2.52409 | 1.8515   | 0.653293 | 0.561441 | -1.52267 | -0.17573 | -1.00597 | 1.186305 | 2.31966  | -3.29979 | 0.107791 | 65.53  | 0     |
| 19 | 2.77122  | -1.55173 | 1.974992 | -0.72332 | 0.413664 | -0.14179 | -1.39934 | -0.04896 | 0.564235 | 0.203365 | 0.820519 | -1.32258 | 2.035967 | -3.00187 | 1.024958 | 320.52 | 0     |

Screenshot B-13: creditcard\_test\_1.csv in Excel — Showing V14–V28, Amount, Class columns

## B-14: XGBoost Model Node — Fields Configuration



Screenshot B-14: XGBoost Tree Node Fields Tab — Target: Class | Excluded: Time, Amount | Predictors: V1–V6... (all engineered features)